# IMO($60^{th}$) – 2019

## First Day

1. Let Z be the set of integers. Determine all functions $f : Z \to Z$ such that, for all integers a and b, $f(2a) + 2f(b) = f(f(a+b))$.
2. In triangle ABC, point $A_1$ lies on side BC and point $B_1$ lies on side AC. Let P and Q be points on segments $AA_1$ and $BB_1$ respectively such that PQ is parallel to AB. Let $P_1$ be a point on line $PB_1$, such that $B_1$ lies strictly between P and $P_1$ and $\angle PP_1C = \angle BAC$. Similarly let $Q_1$ be a point on line $QA_1$ such that $A_1$ lies strictly between Q and $Q_1$ and $\angle CQ_1Q = \angle CBA$. Prove that points $P, Q, P_1, Q_1$ are concyclic.
3. A social network has 2019 users, some pairs of whom are friends. Whenever user A is friend with user B, user B is also friend with user A. Events of the following kind may happen repeatedly, one at a time.

    Three users A, B and C such that A is friend with both B and C, but B and C are not friends, change their friendship statues such that B and C are now friends, but A is no longer friends with B, and no longer friend with C. All other friendship statues are unchanged.

Initially 1010 users have 1009 friends each and 1009 users have 1010 friends each. Prove that there exists a sequence of such events after which each user is friends with atmost one other user.

## Second Day

4. Find all pairs (k, n) of positive integers such that
    $k! = (2^n - 1)(2^n - 2)(2^n - 4)...(2^n - 2^{n-1})$
5. The Bank of Bath issues coins with an H on one side and T on the other. Harry has n of these coins arranged in the line from left to right. He repeatedly performs the following operation: if there are exactly k > 0 coins showing H, then he turns over the $k^{th}$ coin from the left; otherwise, all coins show T and he stops. For example, if n = 3 the process starting with the configuration THT would be $THT \to HHT \to HTT \to TTT$, which stops after three operations.
    (a) Show that for each initial configuration, Harry stops after a finite number of operations.
    (b) For each initial configuration C, let L(C) be the number of operations before Harry stops. For Example, L(THT) = 3 and L(TTT) = 0. Determine the average value of L(C) over all $2^n$ possible initial configurations C.
6. Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle *w* of ABC is tangent to sides BC, CA and AB at D, E and F respectively. The line through D perpendicular to EF meets *w* again at R. The line AR meets *w* again at P. The circumcircles of triangle PCE and PBF meet again at Q.
    Prove that lines DI and PQ meet on the line through A perpendicular to AI.

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